

Putting the “Green” back in Greensburg



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Griffin Statistics
Griffin Impact Challenge



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Outline



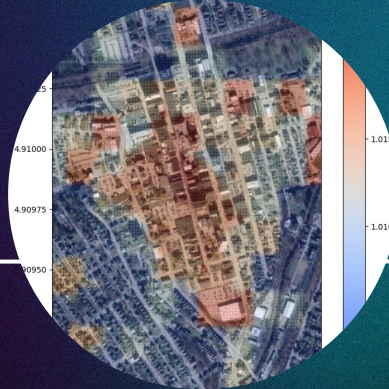
The Urban Heat Island Effect

- ↑ Heat
- ↑ Electricity
- ↓ Money
- ↓ Community Activity



Methodology

- Sentinel - 1 and 2
- Landsat - 8
- LiDAR
- Machine Learning



Findings

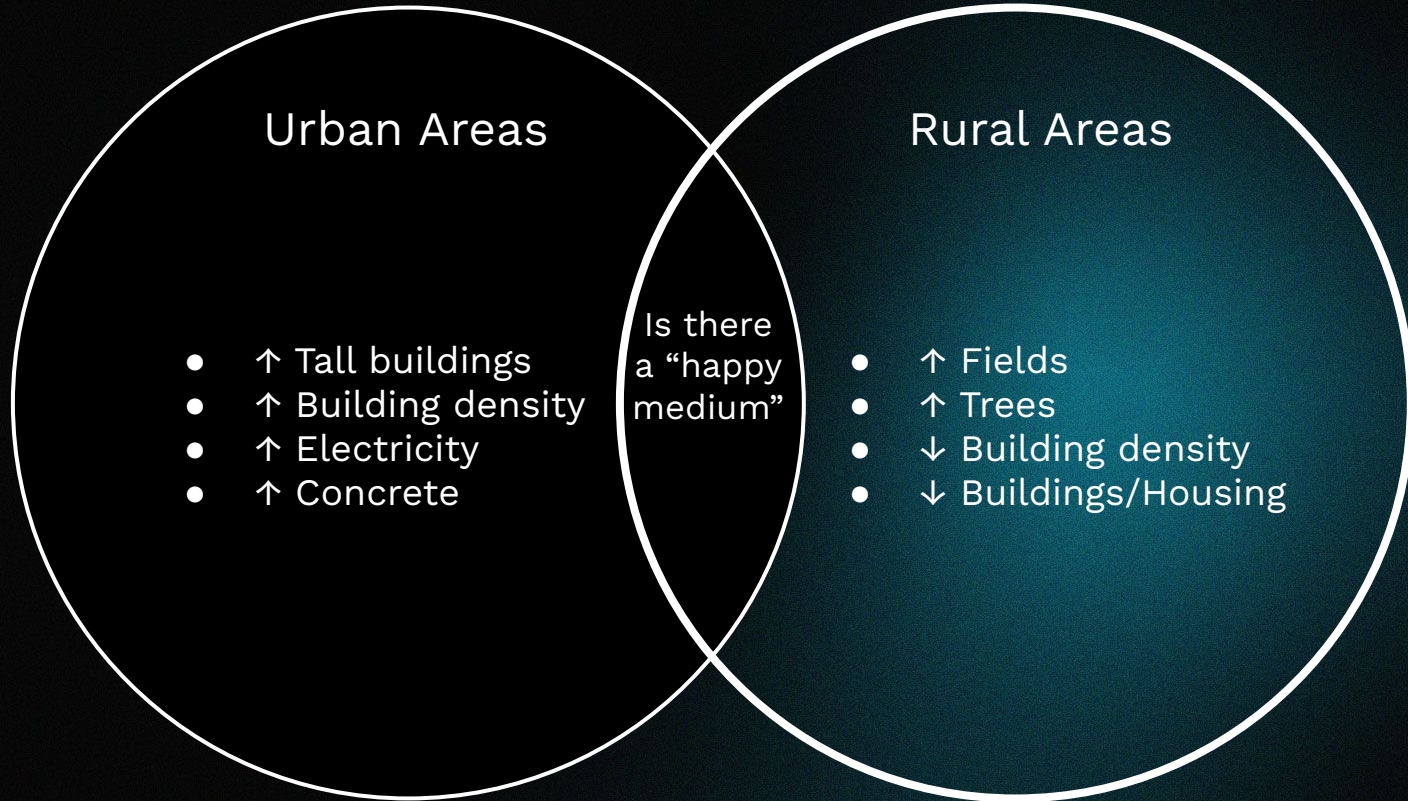
- ↑ City Center Temperatures
- ↓ Suburban Temperatures



Steps For Change

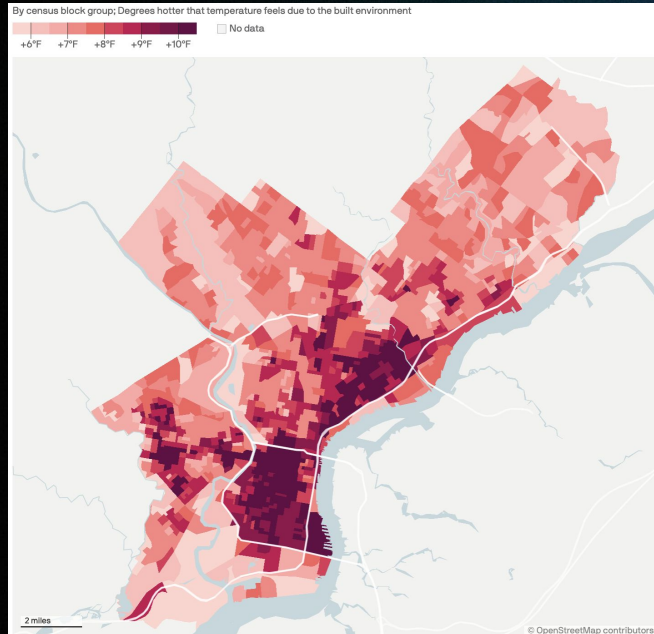
- ↑ Vegetation
- ↑ Vibrancy
- ↑ Community

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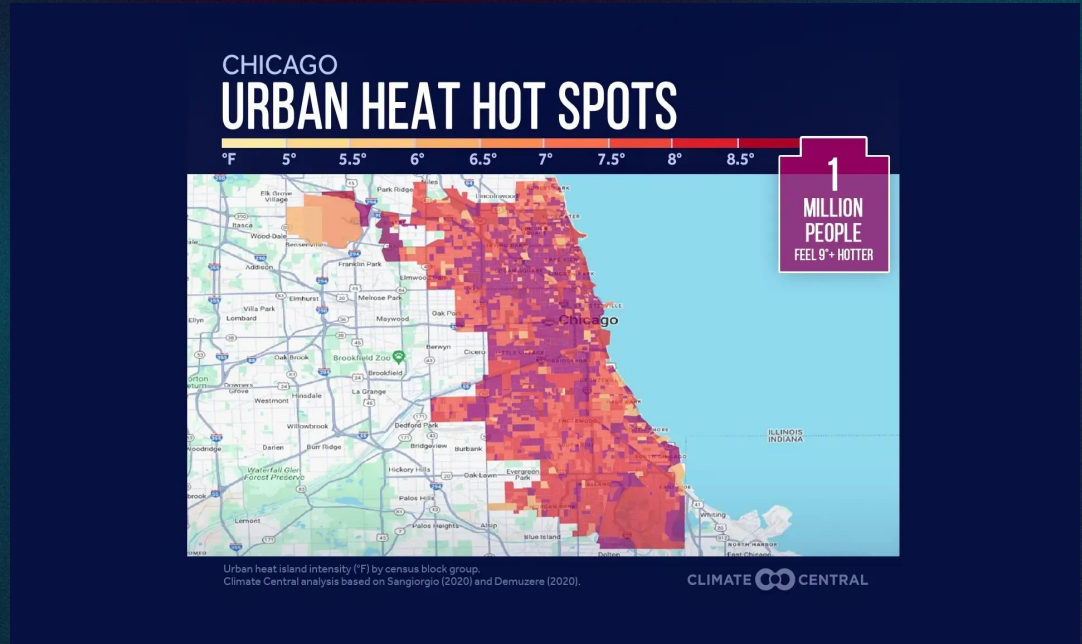


The Urban Heat Island Effect across America

Philadelphia, PA



Chicago, IL



The Costs of Urban Heat Islands



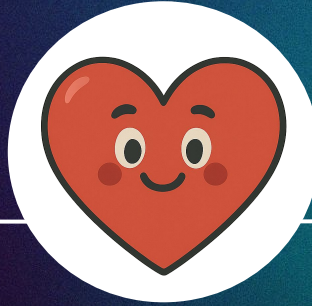
Energy Costs

For a 1.8 degree fahrenheit increase there is a 0.5-5% increase in energy.



Economic Costs

Greensburg companies paying higher electricity bills due to electricity demands.



Health Impacts

Increased risk of health related injuries on warmer day.

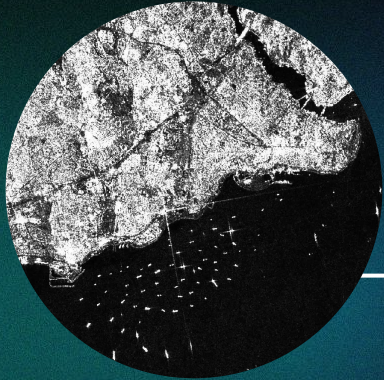


Livability

Less pleasant for public foot traffic.

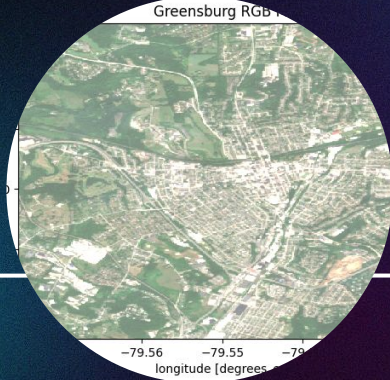
Methodology

July 20, 2024 Data Collected



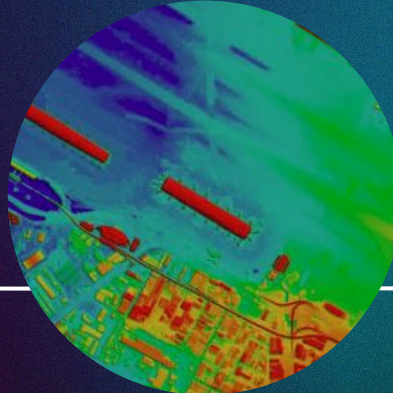
Sentinel - 1 Satellite

- VV
- VH
- Grey Scale Bands



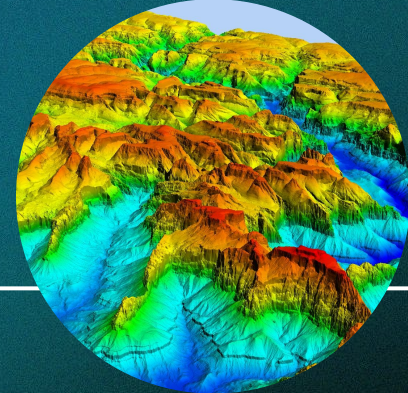
Sentinel - 2 Satellite

- B1 - B12
- B8A
- Color Bands



Landsat - 8 Satellite

- Thermal
- Near Infrared
- Red
- Green
- Blue

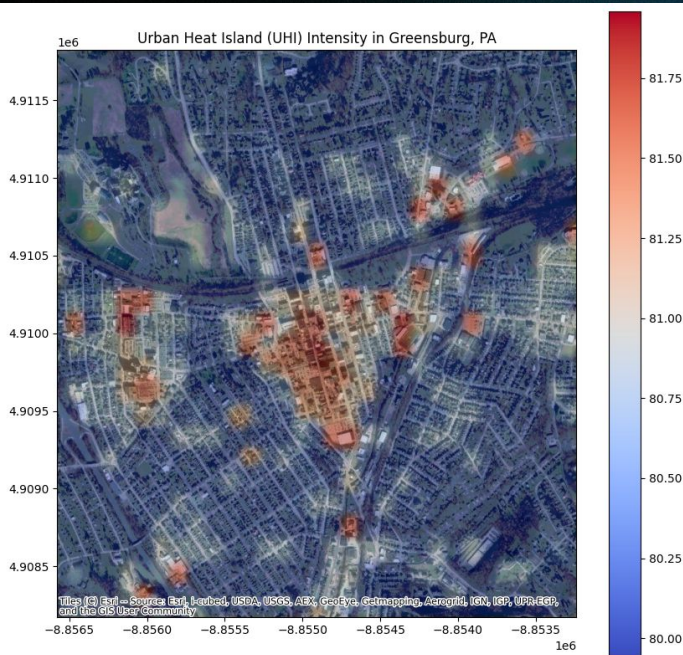


LiDAR Satellite

- DSM (Total Elevation)
- DTM (Natural Elevation)
- HAG (Building Height)

Findings

- Clear indication of Urban Heat Island Effect
- 2 degree fahrenheit increase in downtown Greensburg
- Hottest area is closest to the courthouse



Downtown

Greensburg



Steps For Change



Green Spaces

- ↑ Greenery which improves landscape of downtown
- Offer tax incentives to property owners



Red Maple Trees

- 10\$ for 30 seeds
- Community Planting
- Education Opportunity
- ↑ Tree Canopy Shade



Community Pool

- In-ground swimming pool
- \$50,000 - \$ 100,000
- Community Activity



Roof Solar Panels

- ↑ Electricity
- Renewable Energy
- Dissipates Heat
- \$15,000 to \$35,000 before incentives

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Steps For Change



References

- Kasniza Jumari, N. A., Ahmed, A. N., Huang, Y. F., Ng, J. L., Koo, C. H., Chong, K. L., Sherif, M., & Elshafie, A. (2023). Analysis of urban heat islands with landsat satellite images and GIS in Kuala Lumpur Metropolitan City. *Heliyon*, 9(8). <https://doi.org/10.1016/j.heliyon.2023.e18424>
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